

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 4

Duration: 1 Hour
Total: Marks:40

Annexure A 5

Code of Conduct:

*I (Devadharshini E / 192524483) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: Devadharshini E

INDUSTRY PROBLEM TASK

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

- (a)** Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.
- (b)** Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).
- (c)** Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Diagnosis

- (a)** Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.
- (b)** Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.
- (c)** Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3: Statistical Learning for Risk Stratification

- (a)** Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.
- (b)** Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Treatment Recommendation

- (a)** Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.
- (b)** Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.
- (c)** Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Task 1: Data Preparation & Inductive Learning

(a) Inductive learning approach, target and features

- Inductive learning learns a general rule from labelled examples and applies that rule to unseen patient records. For this problem, supervised binary classification is appropriate because historical patient records contain predictors and a diabetes outcome. A Decision Tree is suitable as an interpretable baseline, while Logistic Regression can provide a statistical risk model.
- Target variable: Diabetes, coded as 1 = diabetic and 0 = non-diabetic.
- Glucose: a major direct indicator of abnormal glucose regulation and therefore an important predictor.
- BMI: higher BMI is associated with increased metabolic and diabetes risk.
- Age: diabetes risk generally increases with age and captures an important demographic effect.
- Blood Pressure: elevated blood pressure is associated with cardiometabolic risk.
- Family History: captures inherited susceptibility and provides additional risk information.

(b) Class imbalance

- If non-diabetic records greatly outnumber diabetic records, a model may obtain high accuracy by mostly predicting the majority class while missing diabetic patients. For this medical use case, recall/sensitivity for the diabetic class is more important than accuracy alone.
- Recommended strategy: use class weights as the primary approach, or SMOTE on the training data when justified. Class weights are attractive because they do not create synthetic patients. If SMOTE is used, it must be applied only to the training set after the train/test split. Aggressive under-sampling is less preferred

because it discards potentially useful patient information.

- Evaluation should report Precision, Recall, F1-score, confusion matrix and AUC-ROC, with particular attention to false negatives.

(c) 70/30 split and preprocessing

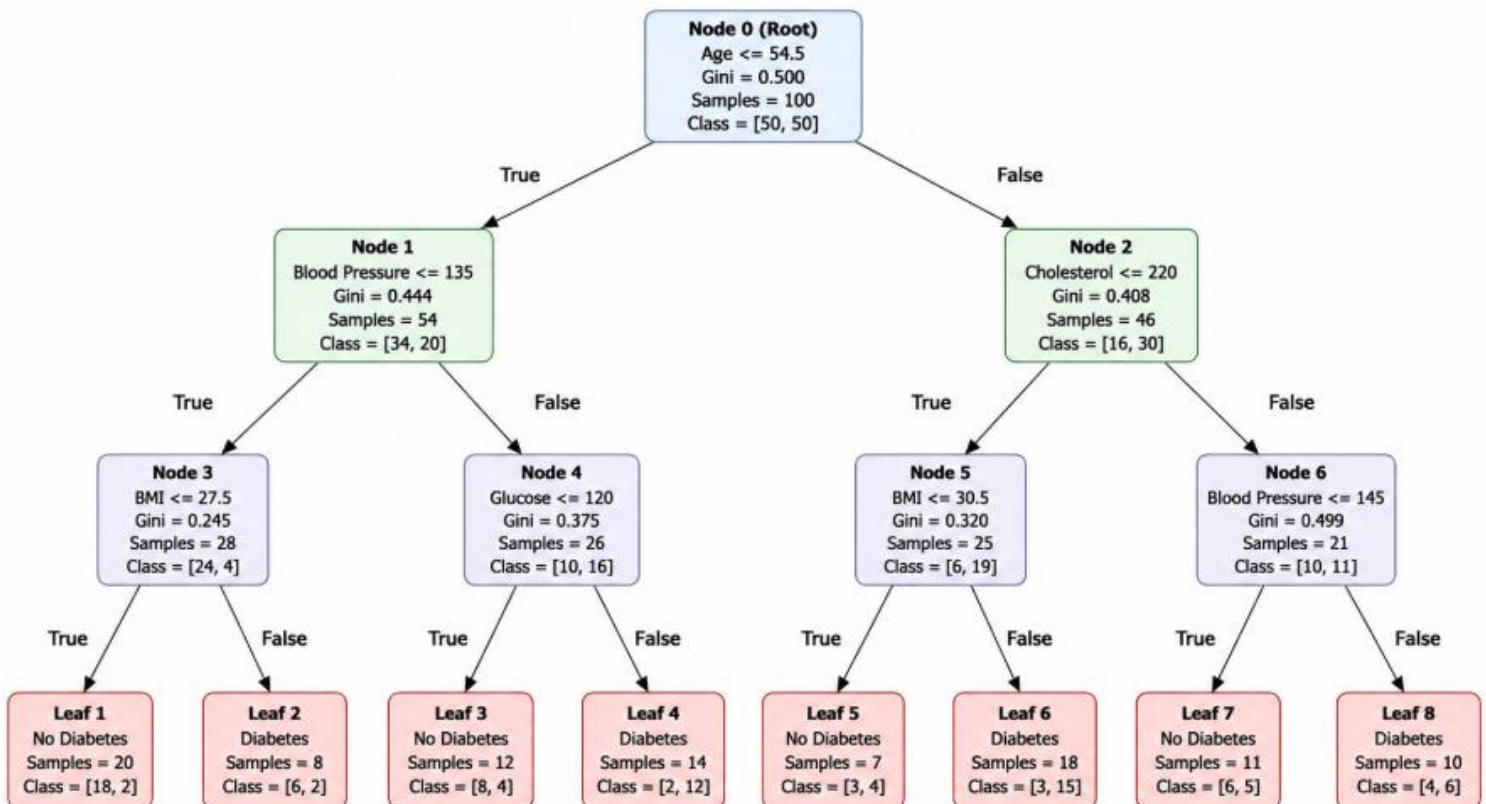
- Use a stratified 70% training and 30% testing split. Stratification keeps approximately the same diabetic/non-diabetic proportion in both subsets. The larger training portion supports model learning, while 30% provides a meaningful independent test set.
- Preprocessing steps:
- Check duplicates, impossible values and missing values.
- Impute missing numeric values using a robust statistic such as the training-set median.
- Encode categorical variables such as family history using binary/one-hot encoding.
- Standardize numeric variables for Logistic Regression or other scale-sensitive algorithms.
- Do not fit preprocessing transformations on the test set; fit them on training data and apply the learned transformation to the test data.
- Perform SMOTE, if selected, only inside the training pipeline to prevent data leakage.
- Impact: normalization improves numerical stability and makes coefficients comparable in linear models. Encoding converts categorical information into machine-readable values. Decision Trees do not require normalization because their splits depend on thresholds rather than distances.

Task 2: Decision Tree for Diagnosis

(a) Tree construction and root selection

- A Decision Tree recursively partitions the training data. At each node, select the feature and threshold that produces the largest impurity reduction. For Gini Index, lower weighted child impurity is preferred. For Information Gain, the split with the greatest reduction in entropy is preferred.
- For an illustrative diabetes dataset, glucose is expected to be a strong root candidate because it can separate high-risk and lower-risk patients effectively. The exact root and thresholds must be calculated from the actual supplied dataset.

Figure 1. Illustrative first three levels of the Decision Tree.



(b) Evaluation and false negatives

Metric	Illustrative value
Accuracy	0.727
Precision	0.314
Recall	0.727
F1	0.438
AUC-ROC	0.766

- Illustrative confusion matrix [TN, FP, FN, TP] = [93, 35, 6, 16]. A false negative is a diabetic patient incorrectly predicted as non-diabetic. In a clinical screening context, false negatives are important because a missed case can delay confirmatory testing and intervention.
- To reduce false negatives: use class weights or SMOTE; lower the classification threshold when probability outputs are available; optimize for recall or a recall-sensitive metric; and send borderline/high-risk cases for clinician review. The threshold must be selected using validation data rather than the test set.

(c) Pre-pruning vs post-pruning

Aspect	Pre-pruning	Post-pruning
When applied	During tree growth	After a larger tree is grown
Typical controls	max_depth, min_samples_leaf, min_impurity_decrease	ccp_alpha / cost-complexity pruning
Main benefit	Prevents excessive growth early	Can remove weak branches after seeing full structure
Explainability	Usually high	Usually high after simplification
Risk	May underfit if too restrictive	May need validation to choose pruning strength

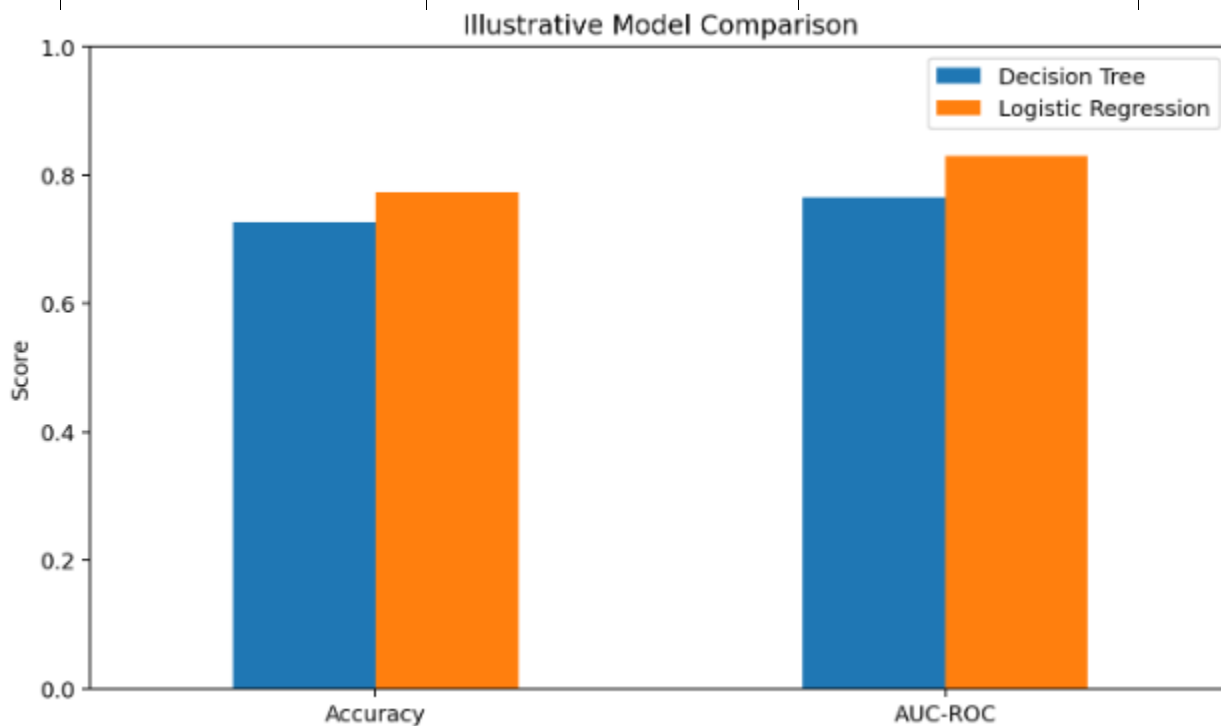
Illustrative post-pruned test accuracy = 0.793, using selected ccp_alpha = 0.000000. For clinical deployment, select the simplest model that maintains acceptable recall and AUC on independent validation data. Explainability and human review are essential.

Task 3: Statistical Learning for Risk Stratification

(a) Logistic Regression and comparison

- Logistic Regression models the probability of diabetes as a function of the predictors. Standardized numeric features are useful because the coefficients then operate on comparable scales. Class weights can compensate for class imbalance.

Model	Accuracy	AUC-ROC
Decision Tree	0.727	0.766
Logistic Regression	0.773	0.830



- A statistical model is preferable when probability/risk estimation, compactness, stable interpretation and approximately linear relationships are important. A Decision Tree may be preferable when nonlinear interactions and rule-like explanations are important. The final clinical choice should be based on external validation, calibration, recall, AUC and interpretability rather than a single metric.

(b) Feature importance

- For the Decision Tree, feature importance is derived from the total impurity reduction contributed by each feature. For Logistic Regression, the absolute standardized coefficient magnitude provides a simple importance ranking.
- Illustrative top predictors:

- Decision Tree: Glucose, BloodPressure, Cholesterol, BMI, Age.
- Logistic Regression: Glucose, BloodPressure, Cholesterol, BMI, FamilyHistory.
- If both models identify similar predictors, that agreement supports the presence of a consistent predictive signal. If rankings differ, the difference can arise from nonlinear relationships, feature interactions or model assumptions. Feature importance does not prove causality.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) MDP formulation

- **States:** Stable, Pre-Diabetic, High-Risk, Diabetic-Uncontrolled and Diabetic-Controlled.
- **Actions:** Diet, Exercise, Medication and Monitor.
- **Reward:** A health-improvement score. Appropriate actions receive positive rewards; inappropriate or unsafe actions receive lower/negative rewards.
- **Transition dynamics:** The chosen action changes the simulated health state. For example, suitable lifestyle actions may move a high-risk state toward pre-diabetic/stable, while an uncontrolled diabetic state may move toward controlled under an effective simulated action.
- **Policy:** A mapping from each state to the action with the highest expected long-term return.
- The treatment MDP is an educational simulation. It must not be interpreted as a real prescription protocol.

(b) Q-learning

- Q-learning updates the action-value function using:
- $Q(s,a) \leftarrow Q(s,a) + \alpha [r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$

- Illustrative parameters: learning rate $\alpha = 0.5$, discount factor $\gamma = 0.9$. Five episodes were simulated, each with three state-action pairs, satisfying the assessment requirement.

Ep.	State	Action	Reward	Next state	Old Q	New Q
1	High-Risk	Diet	3	Pre-Diabetic	0.00	1.50
1	Pre-Diabetic	Exercise	3	Stable	0.00	1.50
1	Stable	Monitor	1	Stable	0.00	0.50
2	Diabetic-Uncontrolled	Medication	4	Diabetic-Controlled	0.00	2.00
2	Diabetic-Controlled	Monitor	3	Diabetic-Controlled	0.00	1.50
2	Diabetic-Controlled	Diet	2	Diabetic-Controlled	0.00	1.68
3	Pre-Diabetic	Diet	3	Stable	0.00	1.73
3	Stable	Exercise	3	Stable	0.00	1.73
3	Stable	Monitor	1	Stable	0.50	1.53
4	High-Risk	Exercise	3	Pre-Diabetic	0.00	2.28

- Convergence criterion: stop when the maximum absolute Q-table change is below a small tolerance such as 0.01 for several consecutive episodes, or when the learned policy remains unchanged for several episodes.

(c) Final learned policy and ethics

- Illustrative learned policy:
- Stable $\rightarrow$ Monitor
- Pre-Diabetic $\rightarrow$ Diet
- High-Risk $\rightarrow$ Exercise
- Diabetic-Uncontrolled $\rightarrow$ Medication
- Diabetic-Controlled $\rightarrow$ Monitor
- Ethical and safety requirements:

- Patient safety: medication-related actions must never be executed autonomously; require clinician authorization and hard safety constraints.
- Bias: assess performance and error rates across relevant patient groups and monitor for unequal false-negative rates.
- Explainability: show the current state, recommended action, key factors and rationale to the clinician.
- Privacy and security: protect patient records, restrict access, log decisions and follow applicable data-protection requirements.
- Human oversight: provide a clinician override, audit trail, monitoring and a rollback procedure.
- Validation: test in simulation and offline historical data first, then conduct appropriate prospective clinical evaluation before deployment.

Overall Conclusion

- The proposed healthcare AI pipeline combines supervised learning for diabetes diagnosis, statistical learning for risk stratification and reinforcement learning as a controlled simulation for treatment-action research.
- The system should prioritize sensitivity to reduce missed diabetic cases, while preserving explainability, fairness, privacy and human clinical oversight.
- The final deployed model should be selected only after validation on the actual dataset and appropriate clinical evaluation.